HazardGuard C3: Multi-Hazard Alarm with a Phone Dashboard

LPG leak, alcohol / solvent vapour, overheating and water-leak detection with traffic-light LEDs and a password-free WiFi dashboard

Category	Level	Main board
Electronics	High school	ESP32-C3

1. ABSTRACT

HazardGuard C3 is a low-cost alarm for kitchens, hotel boiler rooms and workshops. It watches four hazards at once: LPG or natural-gas leaks (MQ-5), alcohol, sanitiser or solvent vapour (MQ-3), overheating or fire (DHT11) and water leaking onto the floor (soil-moisture probe). Three LEDs act like a traffic light: green for safe, a blinking yellow for a warning and a fast red strobe for danger. The ESP32-C3 also creates its own WiFi hotspot, so any phone can open a live dashboard at 192.168.4.1 to see every reading, see which sensors are disconnected, switch between Household, Hotel kitchen and Industrial profiles, test the LEDs and change the alarm limits, which are saved through power cuts.

2. AIM OF THE PROJECT

- Detect gas leaks, solvent vapour, overheating and water leaks with one small board.
- Warn clearly with three LEDs that anyone understands at a glance.
- Show live readings on any phone with no internet and no app.
- Allow different limits for homes, hotels and factories, kept after a power cut.
- Never show a made-up value: a disconnected sensor is reported as 'not connected'.

3. PROBLEM STATEMENT AND NEED

LPG leaks, kitchen fires and burst pipes cause accidents and damage every year, often because nobody notices in time. Most low-cost alarms watch only one hazard, cannot be checked from another room and give no idea whether their sensor is even working. HazardGuard C3 watches four hazards, shows them on any phone nearby and tells you when a sensor has been unplugged.

4. COMPONENTS USED AND COST

Component	Model	Qty	Purpose	Approx. cost
Controller	ESP32-C3 SuperMini	1	Reads sensors, runs logic, WiFi hotspot	Rs. 300
LPG sensor	MQ-5 module	1	LPG / natural-gas leaks	Rs. 150
Alcohol sensor	MQ-3 module	1	Alcohol, sanitiser and solvent vapour	Rs. 150
Temperature sensor	DHT11	1	Overheating / fire, humidity	Rs. 110
Moisture probe	Soil moisture probe + module	1	Water leak / flooding on the floor	Rs. 80
LEDs	Green, yellow, red 5 mm	3	Safe / warning / danger	Rs. 10
Resistors	220 Ω	3	LED current limit	Rs. 5
Power	5 V 2 A USB-C adapter	1	Supply (MQ heaters need 5 V)	Rs. 150
Total				Rs. 955

Table 1: Bill of materials (approximate local prices)

5. CIRCUIT CONNECTIONS

The analog sensors must use the ESP32-C3's ADC1 pins (GPIO 0-4), because ADC2 stops working while WiFi is on. GPIO 2 and GPIO 9 are left empty: they decide how the chip boots, and GPIO 9 is the BOOT button. The MQ outputs go straight to the pins and the gas limits are set for that; the soil module runs on 3.3 V.

Part pin	ESP32-C3 pin	Notes
MQ-5 AO	GPIO 1 (ADC1)	VCC 5 V, DO not used
MQ-3 AO	GPIO 4 (ADC1)	VCC 5 V, DO not used
Soil moisture AO	GPIO 0 (ADC1)	VCC 3.3 V, DO not used
DHT11 DATA	GPIO 6	VCC 3.3 V
Green LED	GPIO 7	220 Ω to GND
Yellow LED	GPIO 8	220 Ω to GND
Red LED	GPIO 3	220 Ω to GND

Table 2: Wiring

6. SYSTEM OVERVIEW

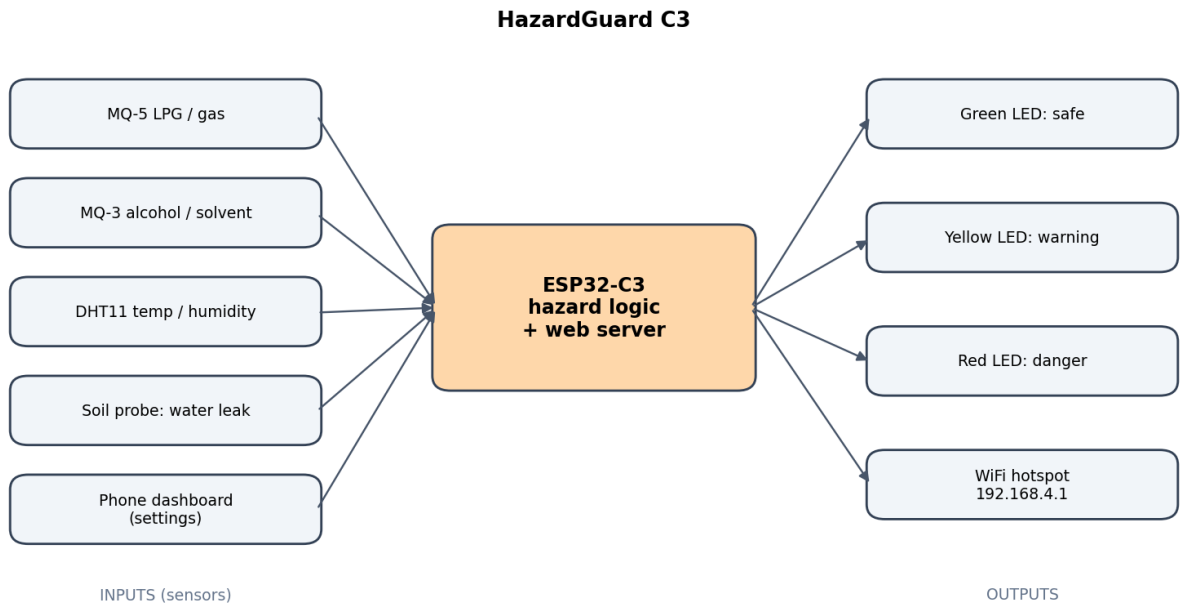


Figure 1: Block diagram

7. HOW I DID THIS (METHOD)

Each analog sensor is read four times a second and smoothed with a moving average of 8 readings. The soil reading is turned into a percentage using two calibration points: the probe in dry air and the probe dipped in water. For the first 60 seconds the MQ heaters warm up, so gas alarms are held back and the yellow LED gives a short blip.

Finding an unplugged sensor is harder than it sounds: a loose analog pin 'floats' and reads random values that look real. Once a second the program switches on the chip's internal pull-down resistor, reads the pin, then the pull-up, and reads again. A wired sensor barely moves, but a floating pin swings from one end of the scale to the other, so it is reported as 'not connected'. The DHT11 is reported after 3 failed readings in a row.

Each profile has its own limits; the Industrial profile is the most sensitive. The chosen profile and any limits edited on the phone are stored in the ESP32's flash memory (Preferences), so they survive power cuts. At power-up the LEDs light green, yellow and red in turn so each one can be checked by eye, and the dashboard can light any LED on demand.

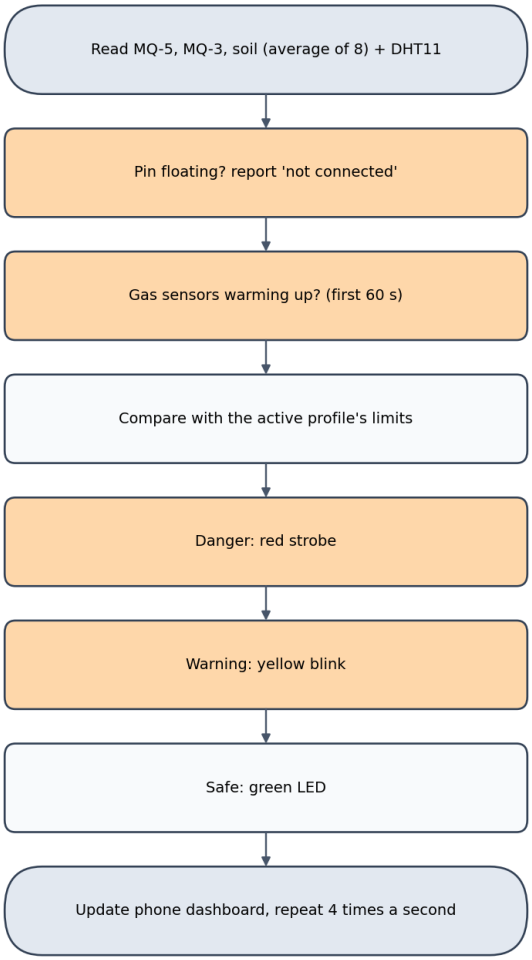


Figure 2: Control flow

8. TESTING AND RESULTS

Figure 3 replays the firmware's decision logic with the Household limits: an LPG leak that is cleared by opening a window, a spilt bottle of sanitiser, and then water spreading across the floor. The bottom strip shows which LED the controller lights at every moment. On the real board the MQ-5 read about 900 in clean room air, far below the 1600 warning limit.

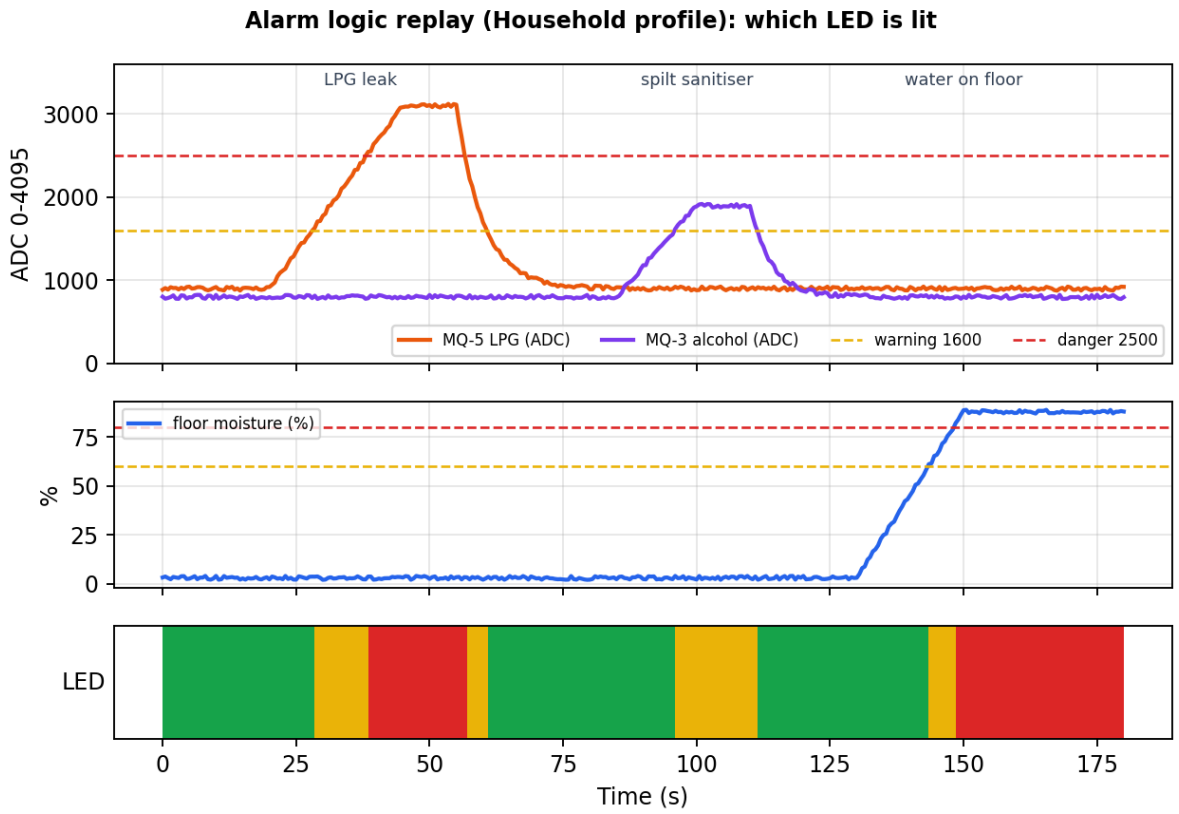


Figure 3: Alarm logic replay

Test on the bench	Expected result
Power on	Green, yellow, red light in turn; then green with a yellow blip during warm-up
Gas lighter (unlit) near the MQ-5	Yellow blink, then red strobe; dashboard: 'LPG / GAS LEAK'
Hand sanitiser near the MQ-3	Yellow, then red; dashboard: 'ALCOHOL / SOLVENT VAPOUR'
Hot air near the DHT11 above 52 °C	Red; dashboard: 'FIRE / OVERHEAT'
Soil probe dipped in a cup of water	Red; dashboard: 'WATER LEAK / FLOODING'
Unplug any sensor	Dashboard lists it as not connected; no fake value

Table 3: Bench test checklist

9. APPLICATIONS

- Home kitchens: LPG leak warning.
- Hotel kitchens and boiler rooms: gas and overheating watch.
- Labs, salons and workshops: solvent and sanitiser vapour.
- Bathrooms, store rooms and server rooms: water-leak alarm.

10. ADVANTAGES AND LIMITATIONS

- Four hazards on one board for under a thousand rupees.
- Works without internet: the phone connects straight to the device.
- Tells you when a sensor is unplugged instead of showing a false 'all clear'.
- Limitation: it only warns; it does not switch off the gas or start a fan.
- Limitation: MQ sensors need warm-up and give relative values, not exact ppm.
- Limitation: the DHT11 reads the air, not flames directly; a flame sensor would react faster.

11. FUTURE IMPROVEMENTS

- Add a relay to start an exhaust fan or close a gas solenoid valve.
- Add a buzzer and a flame sensor.
- Send SMS or app alerts through a home router.
- Battery backup for power cuts.

12. CONCLUSION

HazardGuard C3 watches gas, vapour, heat and water at the same time, shows the result with three simple lights and on any phone, and is honest about sensors that are not working - all for less than a thousand rupees.

13. REFERENCES

- Hanwei Electronics - MQ-3 and MQ-5 gas sensor datasheets.
- Aosong - DHT11 datasheet.
- Espressif - ESP32-C3 technical reference (ADC, GPIO pull resistors) and Arduino core documentation.